

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1570

New Castle 138 kV
November 30, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1570. The Interconnection Customer proposes a 164.9 MW solar photovoltaic facility interconnecting at the New Castle 138 kV substation in the Delaware area. The request was received on September 12, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 356 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,389,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CASTLETON to PLAINFIELD 138 kV
Contingency: BRIDGEPORT SUBSTATION to MARCUS HOOK ENERGY LP 138 kV
Loading: 112.7% (Rating: 337 MVA)
DFAX: 7.3%
Pre-project loading: 109.1%
2. Facility: TRAINER to BRIDGEPORT 138 kV
Contingency: BRIDGEPORT to MARCUS HOOK ENERGY LP 138 kV
Loading: 109.0% (Rating: 331 MVA)
DFAX: 7.3%
Pre-project loading: 105.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CASTLETON to PLAINFIELD 138 kV L	Rebuild line with higher-capacity conductor	\$4,083,000	52% (\$2,125,000)
TRAINER to BRIDGEPORT 138 kV	Reconductor line	\$505,000	52% (\$264,000)
	Total allocated to project		\$2,389,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.